

UNMISH ROY

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EDUCATION

Indian Institute of Information Technology Kalyani

(2022 – Present)

B.Tech in Electronics and Communication Engineering(CGPA:8.05)

WORK EXPERIENCE

Full Stack Web Developer (Intern)

Mar 2025 – Jun 2025

Day and Night Space Foundation – Remote

- Spearheaded the development of a full-stack website using **Next.js**, **FastAPI**, and **CSS**, increasing user engagement by 35% via data-driven design optimizations.
- Collaborated with a cross-functional team of 5+, integrating RESTful APIs and responsive UI for multi-device compatibility.
- Evaluated application performance and optimized load speed by 25% through the implementation of lazy loading, code-splitting, and efficient render strategies.
- Coordinated code reviews and maintained GitHub repositories, ensuring clear communication and timely delivery of project milestones.

TECHNICAL SKILLS

Languages	C, C++, Solidity, Python, JavaScript (ES6+), TypeScript
Frontend	HTML5, CSS3, React.js, Redux, Next.js, Tailwind CSS
Backend & Database	Node.js, Express.js, MongoDB, MySQL
Libraries	Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Mongoose
Generative AI	LangChain, Hugging Face, OpenAI API, Transformers
Dev Tools	Git, GitHub, VS Code, Postman, Docker, Cloudinary

PROJECTS

AI Virtual Assistant with Voice Command Integration

[\[GitHub Link\]](#)

- Built a full-stack AI-powered assistant using the **MERN stack** and **Cloudinary** for Organised media handling.
- Developed and implemented secure **authentication & authorization** mechanisms (JWT, bcrypt), enabling reliable role-based access control.
- Designed and deployed an **ML-based speech-to-text and natural language understanding** pipeline delivering 95% accuracy in real-time voice command execution, reducing manual input tasks.
- Coordinated with peers for API integration, UI consistency, and comprehensive documentation.
- Accelerated performance via asynchronous data fetching & state management, which improved response speed.

Anomaly Detection in Epilepsy Using EEG Signals

[\[GitHub Link\]](#)

- Designed and implemented a deep learning pipeline using **1D CNN**, **LSTM**, and **GRU** models to classify epileptic episodes with 97% accuracy.
- Processed and analyzed 5,000+ EEG records with segmentation, normalization, and noise reduction, improving signal clarity by 40%.
- Collaborated in a 3-member research team to integrate advanced feature extraction and multi-metric model validation.
- Evaluated performance using confusion matrix, ROC curves, and stratified k-fold cross-validation, ensuring high reliability in predictions.

Parkinson's Disease Detection using ANN & Hybrid Model (LSTM + Time Series)

[\[GitHub Link\]](#)

- Built deep learning healthcare models using **ANN** and a **hybrid LSTM + time series** architecture for early Parkinson's detection, achieving over 93% accuracy.
- Conducted feature engineering, normalization, and dimensionality reduction on 10,000+ biomedical data points, improving model training efficiency by 30%.
- Trained and optimized models using **Python**, **TensorFlow**, and **scikit-learn** to enhance generalization.
- Measured outcomes via accuracy, precision, recall, and F1-score, validating clinical applicability.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- **Leadership:** Led and executed a high-impact innovation event at IIIT Kalyani, managing a 5-member core team and collaborating with GIAN, HBNCRIIA, and IIC to host 400+ participants.
- **Sports:** Represented IIIT Kalyani at the 6th Inter-IIIT Sports Meet (Football), showcasing teamwork, strategy, and collaboration.